

Pre-processing and exploratory data analysis

1. Data preprocessing

Some data preprocessing steps are done before initial data analysis for data cleaning and exploration purposes.

1) Generating new features

Try to generate several new features here for broader exploration possibilities. Since there is a datetime feature in the data, two more features are generated: day of the week and a bool column indicating if it's a weekday, to further explore the relation between border-crossing behavior and weekend/weekdays

day_of_the_week	is_week_day
4	1
4	1
4	1
4	1
4	1

2) Joining more data

The original dataset is purely focused on border-crossing stats such as port information and crossing measurement. In order to excavate extra info, two datasets are joined: the GDP stats of each state, and the average temperature of each state, to explore the relation between border-crossing behavior and economical/thermal state.

state	2005:Q1	2005:Q2	2005:Q3	2005:Q4	2006:Q1	2006:Q2	2006:Q3	2006:Q4	2007:Q1	2007:Q2	2007:Q3	2007:Q4	2008:Q1	2008:Q2	2008:Q3
Alaska	37377.7	38626	40251.7	42843.8	42633.9	44349.8	45038.2	45719.4	46441.9	48737.4	49602.3	51108.7	52412.2	55933.6	57385.1
Arizona	219858.1	225360.6	232317	235388.9	242446.7	245035.7	250166.7	255273.5	257952	261341.6	265246	264748.5	262590.9	261815.6	25992.1
California	1711573.3	1734383.7	1771679.5	1797828.4	1852200	1859177.9	1883269.3	1904301.9	1916188.9	1940770.5	1966137.7	2000327	1971753.2	2012750.2	20198.1
Idaho	46775.3	47564.9	48675.8	48835.6	50297.2	51074.4	50734.7	51798.8	52797.7	53365.4	54058.5	55186.7	54958.8	55999.5	56171.1
Maine	45234.7	45527.7	45969.6	46767.3	47242.6	47902.8	48108.7	48430.3	48385.9	49154.2	49475.5	49578.8	49706.9	49975.6	50279.1
Michigan	392832.7	394652.3	393789.4	395641.6	392521.3	395820.3	396592.3	396870.2	404282.4	405322.2	406305.7	399032.8	390804.1	388781.9	38405.1
Minnesota	238803.4	240523.2	246503.5	245620.7	247639.3	251120.5	249202.5	250977.5	253486.8	258973.1	260097.9	262751	263548.6	264061.2	26861.1
Montana	29759.4	29951.1	30700.8	31707.5	31827.7	32528.3	33401.6	33975.6	35190.4	36003.3	36274.5	36828.1	36833.9	37196.1	37212.1
New Mexico	72148.1	73152	73994.8	75441.7	76771.8	77172.1	77569.5	78191.1	78518.9	80243.2	81583.5	81945.6	81702.1	84274.2	84911.1

GDP by state dataset

Date	Value	Anomaly
199601	24.0	1.9
199602	27.0	0.2
199603	34.8	2.0
199604	41.4	0.3
199605	46.5	-3.0
199606	58.3	1.5
199607	67.0	1.7
199608	64.5	0.8
199609	52.5	-2.4
199610	43.4	-0.8
199611	33.4	1.9
199612	25.3	1.1

Average temperature by state dataset

3) Standardization

For numeric features such as geographical coordinate of ports, border crossing value, etc., they are standardized to facilitate further modeling process

4) One-hot encoding

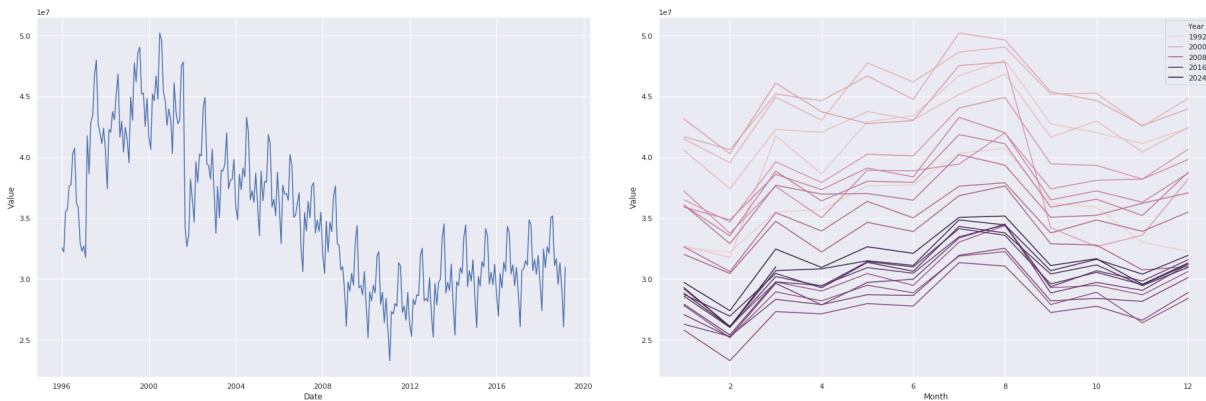
For categorical features such as state, measurement and border type(US-Canada or US-Mexico), etc., they are one-hot encoded to multiple columns so that they can be further fed into models that only accept numerical inputs.

2. Exploratory data analysis

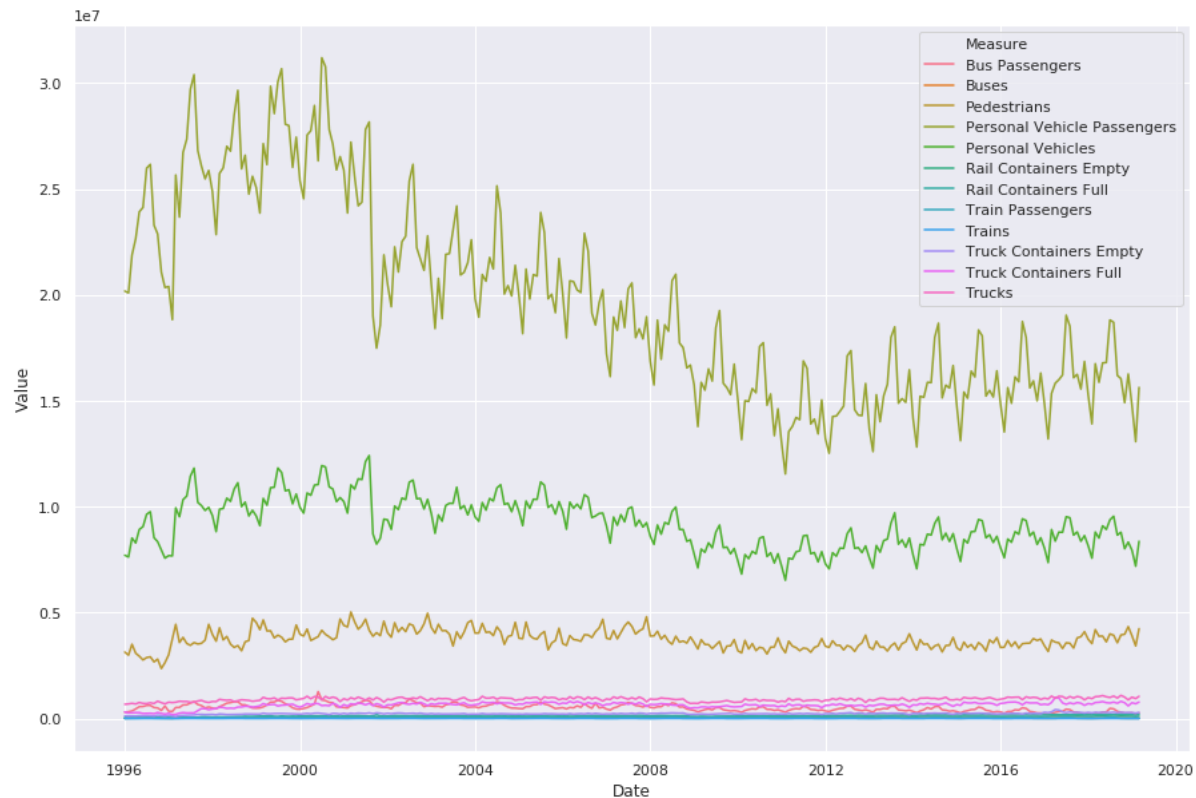
Some exploratory data analysis are done for initial data pattern exploration, as well as for data wrangling before feeding it into further data science models.

1) explorative visualization

Below is the line charts showing the overall border-crossing quantities going over time, as well as month breakdowns.



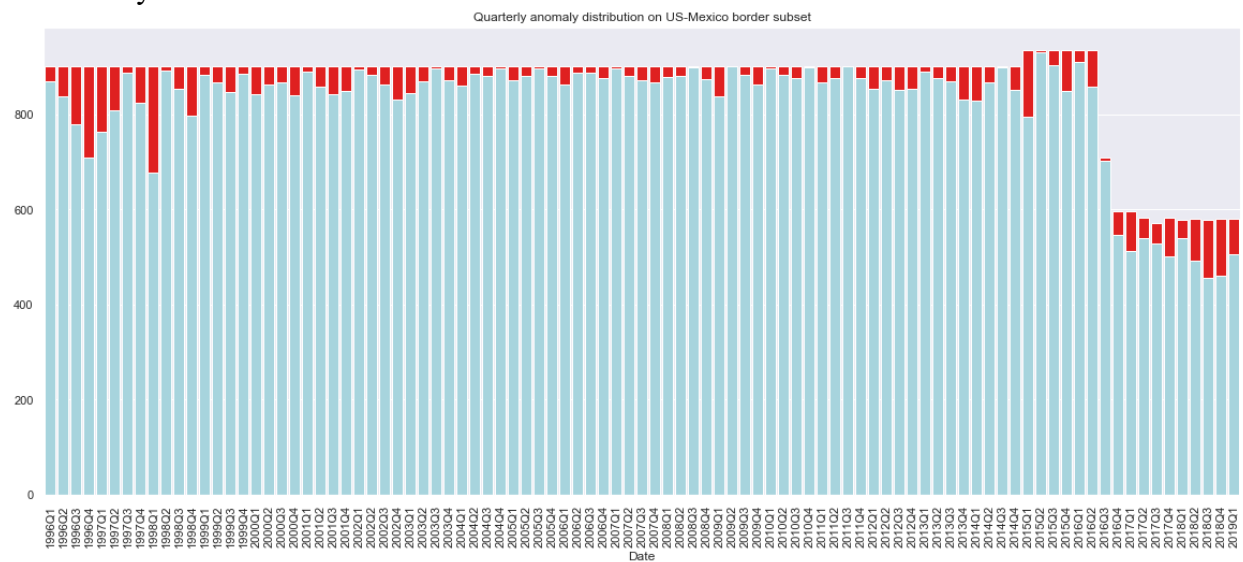
And the aggregated quantities by measure (e.g. by truck/train/vehicle, etc.)



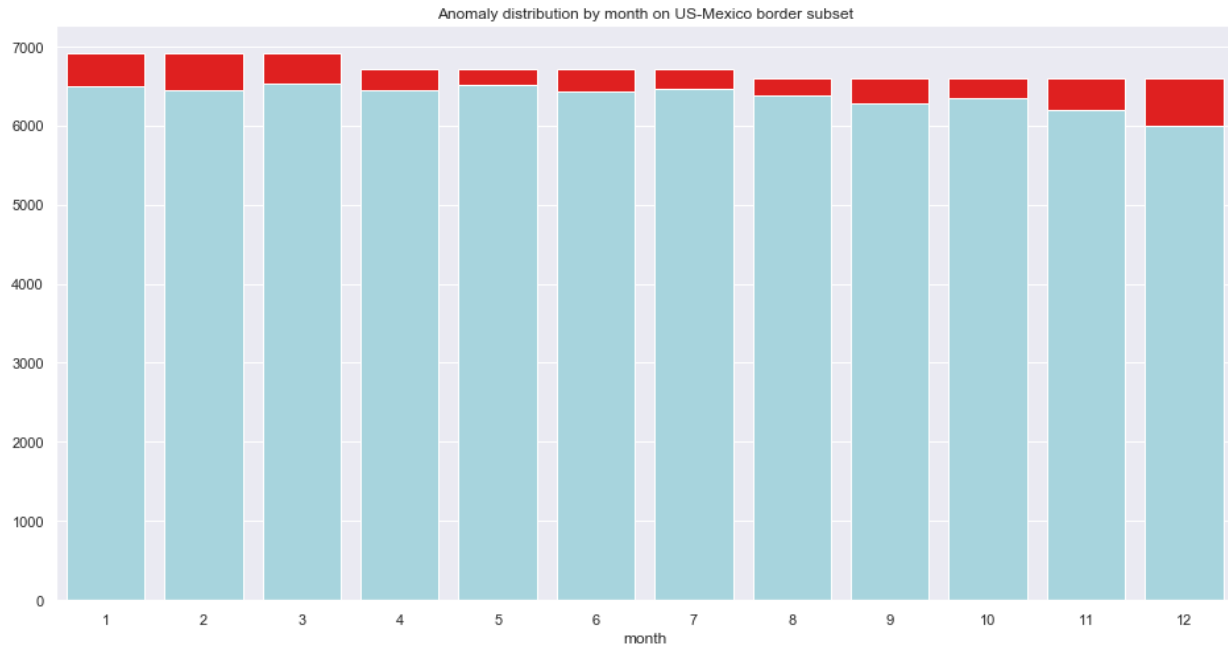
A strong seasonal trend could be observed in the above plots, which indicates that border-crossing behaviors are somewhat related with seasonality and temperature.

2) Anomaly detection

Tried several machine learning algorithms here to detect anomalies, such as isolation forest and one-class SVM. The below visualization shows the quarterly outlier distribution detected by isolation forest.

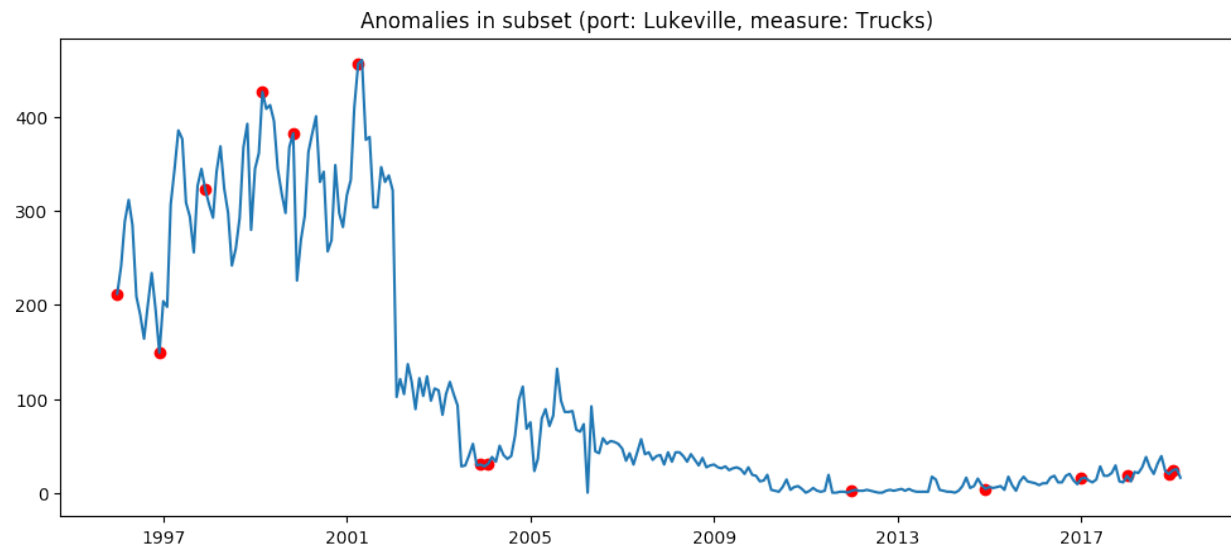


And below illustrates the outlier distribution by month overall, detected by isolation forest.



It could be observed that outliers are more prone to happen before 1998 Q3 and after 2017 Q4. Looking by month, there seems to be more outliers in December, January and February.

Taking a closer look at a specific port (Lukeville) with a specific measure (Trucks). One-class SVM detects the outlier pattern as below.



Outliers detected are mainly distributed on data peaks and troughs, and evenly distributed across all years in the dataset.